

Q24 Ans: D

Inserting an iron core concentrates the magnetic field lines and provides better linkage between coils P and Q. Hence the flux linkage through both coils will be larger and amplitude of the induced voltage in coil Q will increase.

Option A is incorrect – although increasing the frequency will increase the amplitude in coil Q (due to a faster rate of change of flux linkage), the number of cycles per unit time (i.e. frequency) will also increase.

Option B is incorrect – decreasing the cross-sectional area of coil P will have no effect on the flux density created by coil P. Hence no change to the flux linkage of both coils.



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Option C is incorrect – decreasing the number of turns in coil Q will result in a lower amplitude as

$$\frac{V_Q}{V_P} = \frac{N_Q}{N_P}.$$

Q25 Ans: C